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Kedist Shegute Dimore

AI Developer &
Knowledge Graph Engineer

Contact

Email
kedistkid723@gmail.com
Phone
+971 52 177 5993
+251 901 946 736
Location
Addis Ababa, Ethiopia
GitHub
github.com/kedistS
LinkedIn
linkedin.com/in/kedist-shegute

Skills

Languages
Python · Java · C++ · JavaScript · Dart · PHP

AI & Graph
TensorFlow · Keras · Neo4j · Graph mining

Web & Apps
FastAPI · React · Flutter · Node.js · Bootstrap · Tailwind

Data & DevOps
PostgreSQL · MySQL · MongoDB · Docker

Profile

AI Developer and Computer Scientist with 3+ years of professional experience building intelligent, data-driven systems at iCog Labs. Contributed to one of the largest biomedical knowledge graphs in active research, analyzed recurrent graph patterns in BioCypher KG, TFLink, and STRING PPI networks, and led a multi-service AI platform from design to deployment. Equally comfortable across machine learning, backend engineering, and full-stack development. Licensed software developer in the UAE.

Experience

Team Leader, NeuroGraph AI Assistant

iCog Labs | October 2025 to April 2026

- Authored the project proposal and technical framework for NeuroGraph, a system for neural subgraph mining, explanation, and visualization, then led the team that built it.
- Led three core components: a neural subgraph pattern mining engine inspired by SPMiner-style workflows, an LLM-powered interpreter, and an interactive visualization interface for querying motifs across graphs.
- Coordinated four independent services, managed code reviews and Docker production deployment, and kept the team aligned from proposal through delivery.

Computer Scientist

iCog Labs | April 2023 to Present

- Helped build and maintain a large-scale biomedical knowledge graph integrating 34 human data sources and 11 Drosophila sources, spanning millions of nodes and edges across genes, proteins, regulatory elements, and diseases.
- Integrated genomics, proteomics, and disease biology data while keeping structure, relationships, and refreshes reliable across the graph.
- Redesigned the data import pipeline to load large graph updates significantly faster, enabling more frequent and reliable refreshes.
- Built an automated source-release detection and update system, reducing manual maintenance effort.

Graph Pattern Mining & Biological Discovery

iCog Labs / Rejuve.Bio

- Applied neural subgraph mining to a BioCypher human biological KG with 150,619 nodes and approximately 20M edges, sampling 20,000 ego-network neighborhoods.
- Characterized 18 patterns across 11,311 discovered KG instances and 10,138 unique entities, separating dense PPI subgraphs, flexible annotation fans, and cross-domain motifs.
- Analyzed TFLink and STRING follow-up results, including 503 recurrent regulatory pattern instances and 14,877 STRING PPI pattern instances across 18 motif types.

Impact

20M edge BioCypher KG analyzed

11,311 KG pattern instances discovered

14,877 STRING PPI pattern instances analyzed

503 TFLink regulatory motifs reported

License

Software Developer
Creative Media Authority
Abu Dhabi, UAE
License #3767
Valid: Oct 2025 - Oct 2026

Languages

English	Advanced
Amharic	Native
Korean	Reading/listening

Interests

- Artificial Intelligence
- Graph Science
- Web Development
- Technical Writing

Tools

Engineering
Git · Linux · Selenium · WordPress · CI/CD

Graph & Bio
Neo4j GDS · SPMiner · BioCypher · NetworkX · OWL/RDF

Engineering

Engineering & Automation

- Built affected-area CI/CD testing, generated schema documentation from source code, and containerized the full AI assistant platform using Docker.

Personal Projects

Coffee Disease Detection and Classification System

Final Year Project | 2023 to 2024

Built an end-to-end ML pipeline for coffee plant disease detection, from data collection and labeling through model training, evaluation, API integration, and frontend delivery.

Campus PC Authentication System

Personal Project | 2023

Built a Flutter mobile app for university device registration, generating QR code tags that security staff scan to verify device ownership.

Lounge Management System

Personal Project | 2022 to 2023

Developed a Java Swing desktop app for campus lounge inventory, customer ordering, and real-time order tracking.

Research

Neural Subgraph Mining Analysis of a Human Biological Knowledge Graph

Analysis Report | BioCypher KG, 2026

Analyzed a 150,619-node, ~20M-edge KG integrating GO, STRING, TFLink, GENCODE, UniProt, and GAF data; characterized 11,311 discovered instances across 18 motifs.

Hierarchical Regulatory Patterns in Human Transcription Factor Networks

Manuscript | iCog Labs / Rejuve.Bio, 2026

Analyzed the TFLink network and reported 503 recurrent hierarchical pattern instances across 374 unique genes, including AGPAT5 and IMPACT as frequent terminal convergence points.

Structural Motif Analysis in Protein-Protein Interaction Networks

Manuscript | iCog Labs / Rejuve.Bio, 2026

Analyzed STRING PPI mining results with 14,877 pattern instances across 18 motif types, highlighting modules in immune signaling, chromatin regulation, mitochondria, DNA replication, and transport.

Education

B.Sc. in Computer Science

Graduated: July 2024

Certifications

Coursera Specializations

- Supervised Machine Learning: Regression and Classification
- Neural Networks and Deep Learning
- Mathematics for Machine Learning: Linear Algebra
- Object-Oriented Design